Structural Analysis & RCC Design Fundamentals

Department: Civil Engineering | Semester: 4 | Code: CE402
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1. Shear Force & Bending Moment Diagrams

Shear force (V) is the algebraic sum of vertical forces acting on either side of a beam section. Bending moment (M) is the algebraic sum of moments of forces about that section: dM/dx = V.

2. Limit State Design for Concrete Structures

According to IS 456:2000, structures must satisfy limit states of collapse (flexure, compression, shear, torsion) and serviceability (deflection, cracking).

3. Soil Mechanics & Shallow Foundations

Terzaghi's bearing capacity equation computes ultimate bearing capacity based on soil cohesion, surcharge depth, and unit weight.